Mobile Development
Messaging/Background Jobs

LAB 11/W13

3ND YEAR - ENSIA

IMPORTANT

- If you are found to use any LLM (including copilote) inside the lab and you are warned three times, expect **zero mark for your CC**.

OBJECTIVES

- Learning how to integrate messaging/notification to increase user engagement.
- Practice simple use to learn how to use Background jobs.

INDIVIDUAL EXERCISES

Task / Exercise C1 (Creating a Cron Job) [45 minutes + 15 minutes discussion]:

This exercise must be tested only on Android Devices (Not chrome/Linux/Windows)

Download the following project :
https://drive.google.com/file/d/1GnjrpmOdpGA-dA1fO479KucDCxFtToZf/view?usp=drive_link

Implement the following :

1. For the Counter Example, create a **cron job** that increments the counter by every **2 minutes** even when the app is active. // See the slides for Code (NOT CHAT GPT)
   a. Does it work when the App is not in the Foreground ?

2. There should be another background task to reset the counter to zero. At some specific time, for instance **at every hour** at 15th minute, for instance, 11:15, 12:15..,
   Important :
   a. **it** must work/run even when the App is **closed.**
   b. Can you call the Cubit from a background task when the App is closed ?

2nd TEAM CHALLENGE

Task / Exercise C2 (Firebase Messaging) [120 minutes + demonstration on the go]:

Students should progress on their projects to integrate Messaging/notifications where :

- Simple Backend is developed to initiate sending custom messages to some mobile app users
- Targeted Users should be able to receive the messages when the app is closed or open
- Upon clicking on the notification message box, users should be taken to the correct screen (Be creative to choose whichever screen is relevant)..
- Optionally:
  - Received messages should be stored locally for the user to read at a later time.
  - The backend should initiate sending messaging automatically, periodically at some specified times/durations.

Marking :

- Team members should demonstrate the messaging and notification on real phones or emulators.

Technology :

- No restriction on the technology, use whichever technology is relevant to you.

Lecture Demonstration Files:

https://drive.google.com/file/d/1O6AcCuEMnzrLxb1ZW5RYr_OV6LzQ_QrI/view?usp=sharing